

Student / Shop Handout — Recommended Tool & Equipment List

From program Appendix A (expanded for delivery). Quantities assume ideal class of 6–8.

Diagnostic & electrical

Item	Spec / notes	Qty guidance
Digital multimeter	True RMS, CAT III 600V min (Fluke 87V or equiv.)	1 per 1–2 students
DC clamp meter	Preferred for charger/motor current	1–2
Battery load tester	100–200A carbon pile or electronic	1–2
Hydrometer or digital refractometer	With temperature compensation chart	1 per team
Thermometer / IR temp gun	For SG compensation & connector heat	1–2
Insulation tester (megger)	Optional; isolate motor from controller	1
Curtis/Alltrax programmer	Optional but valuable	1

Hand tools & repair

Item	Notes
Insulated screwdrivers, pliers, wrenches	Inspect insulation regularly
Torque wrench set	in-lb and ft-lb
Ratcheting wire crimper + terminals + heat shrink	Training consumables stocked
Soldering iron/station + rosin-core solder	With stand
Dielectric grease, terminal protectant, contact cleaner	
Battery terminal brush / neutralize supplies	Baking soda, rinse water
Jack + rated jack stands + chocks	Mechanical week
Compression tester + safe spark tester	Gas week

PPE & emergency

Item	Notes
Safety glasses	Every student
Face shields	Battery work
Insulated gloves Class 00 + leather over-gloves	Sized pairs
Shop aprons, steel-toe boots policy	
Fire extinguisher Class C or ABC	Charging & gas areas
Eyewash station, spill kit	Tagged & inspected

Information resources

Item	Notes
OEM service manuals / digital access	Club Car, EZ-GO, Yamaha
Wiring diagram display	Poster or large screen
Curtis fault code access	Print + https://faultcodes.curtisinstruments.com/

Training aids

Item	Notes
4-6 golf carts	Mix 36/48V LA, ≥1 lithium, 1-2 gas
Electrical training boards	Week 1
Cutaway battery or photo set	Week 2
Planted-fault kits	Fused short box, labeled opens — never uncontrolled pack shorts